

# Watermarking Algorithm with Low Density Parity Check Codes and Watermark Estimation<sup>1</sup>

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## Abstract

The goal of this paper is to improve the watermarking performance using the following two methods; watermark estimation and low density parity check (LDPC) codes. For a blind watermark decoding, the power of a host image, which is hundreds times greater than the watermark power, is the main noise source to detector. Therefore, a technique that reduces the effect of the power of the host image to the detector is required. To this end, we need to estimate watermark from the watermarked image. With the generalized Gaussian distribution modeling of sub-band coefficients in the wavelet domain, the watermark estimation can be done by an adaptive thresholding estimation method for image de-noising. The watermark capacity as well as the error rate can be improved by applying optimum decoding principles and error correcting codes (ECC). In this paper, we exploit error correction technique using LDPC codes for the decoding of the estimated watermark. Since the exact embedded watermark power is known, the channel noise power can be calculated from the estimated watermark. Note that in LDPC codes, the knowledge of the exact noise power improves the error correction capability. Simulations results demonstrate the superior performance of the proposed algorithm comparing to LDPC decoding of the estimated watermark with assumed channel noise power.

## 1. Introduction

Blind digital watermarking is the art of communicating a message by embedding it into multimedia data (host data), and decoding it without access to the original non-watermarked host data. In general, blind watermarking schemes were built on the principle of spread spectrum. Although many experiments have been performed successfully upon blind watermark decoding whose performance, in ideal situation, is equivalent to the decoding with a host image, its performance cannot be guaranteed because the blind detection of spread-spectrum watermarks suffers significantly from host data interference [1]. Therefore, a technique which reduces the effect of the power of the host image to the detector is required. This can be accomplished by a whitening process, which also guarantees reliable communication. That is, to improve the watermark performance we need to estimate the original watermark as accurate as possible. In this paper, a given watermarked image, we apply an efficient de-noising technique to estimate the watermark. Since the added watermark can be considered as a noise, image de-noising scheme can be used to separate the additive noise from the host signal. In recent years there has been a fair

amount of research on wavelet thresholding and threshold selection for signal de-noising because wavelet provides an appropriate basis for separating noise signal from the noisy image signal [2]. Since the wavelet transform has a good energy compaction property, the small coefficients are more likely from noise (added watermark) and large coefficients from the original signal. These small coefficients can be thresholded without significant degradation of the image. Thresholding is a simple non-linear technique, which operates on one wavelet coefficient at a time. In this paper, replacing the large signal coefficients by zero and taking inverse wavelet, we have a watermark signal with lower image power.

LDPC codes [3] are a class of binary linear block codes, which have parity check matrices that consist mainly of 0 and only a small number of 1, i.e. they are very sparse. An  $(n, i, j)$  LDPC code is defined to have a block length  $n$  and a parity check matrix with exactly  $i$  1's per column and  $j$  1's per row. Like turbo codes, LDPC codes can achieve near Shannon limit error performance and represent a very promising prospect for error control coding [4]. To get this best performance of LDPC codes, some conditions are required. One of them is that we need to know the exact channel noise power, which is rarely

available. Thus, there will be mismatch between the assumed noise power and the true one [4]. However exploiting the exact knowledge of the embedded watermark power, we can calculate the channel noise power from the estimated watermark.

In this paper, a whitening process is adopted to estimate the watermark in the wavelet domain. Then, we can improve the decoding performance of the estimated watermark using LDPC codes with the exact channel noise power calculated from the estimated watermark.

The rest of this paper is organized as follows. The next section covers the proposed watermark embedding process which consists of message assignment and Human Visual System (HVS). In section 3, we explain the watermark estimation in the wavelet domain. Section 4 is devoted to message encoding and decoding with LDPC codes. Simulation results are given in section 5. Finally, the main contributions of this work are summarized in section 6.

## 2. Watermark Embedding Process

Figure 1 shows the overall diagram of the proposed watermarking algorithm, which operates in the spatial domain. Embedding process consists of three steps, namely message assignment with M-ary modulation, message encoding with LDPC codes, and watermark embedding with the Human Visual System (HVS). Detection process includes watermark estimation, message decoding with LDPC codes, and message extraction with M-ary demodulation.

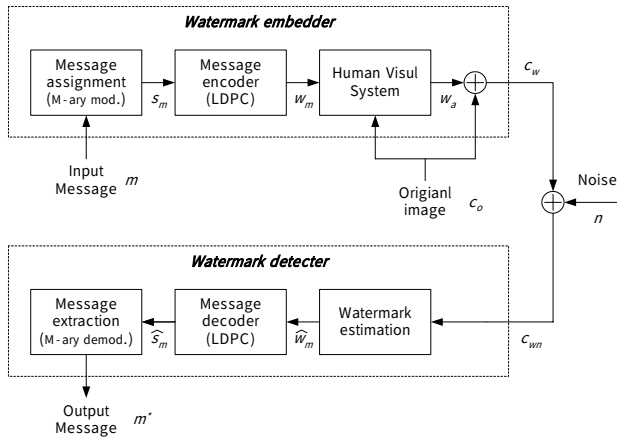


Fig. 1 Diagram of the proposed watermarking algorithm

### 2.1 Message Assignment

This step is to assign a unique and predefined message sequence to each input message. We assume that an image is partitioned into  $B$  sub-image blocks. We suppose that a block has a size of  $R$ . We propose to use M-ary modulation to embed  $A$  bits [5]. Given  $B$  blocks, one character ( $A/B$  bits) is embedded into each block. For brevity, we explain only for one block, the other blocks are allocated in the same manner.

### Array allocation procedure

- A. Assume that one character is embedded to each block.
- B. We generate the message sequence vectors  $s_m \in \{0,1\}$  having a length  $L$  ( $L < R$ ) from the seed (input) message  $m \in \{1, 2, \dots, 2^{A/B}\}$ .
- C. Any two sequences are mutually orthogonal.
- D. The value  $m$  represents the message of the corresponding block.

After assigning the message sequences, a small amount of controlled redundancy is added to produce a codeword  $w_m$ , which results in an error correction code (ECC). The size of codeword is same as the one of the partitioned block  $R$ . The goal of ECC is to maximize the number of errors that can be corrected while introducing as little redundancy as possible. In this paper, we increase the watermark hosting capacity of still images by exploiting the ECC, low density parity check (LDPC) codes [4] in particular. They are known to be best performing to the very near to the Shannon limits [3]. The detail of LDPC is described in section 4.

### 2.2 Human Visual System

Let  $c_o$  be the luminance value of an original image. Then, the embedding process can be represented as follows;

$$c_w = c_o + \alpha w_m = c_o + w_a, \quad (1)$$

where  $w_m$  is the LDPC coded sequence that is a random variable with a *Bernoulli* ( $p$ ) distribution such as

$$w_m = \begin{cases} +1 & \text{with probability } p \\ -1 & \text{with probability } 1-p \end{cases} \quad 0 \leq p \leq 1 \quad (2)$$

and the parameter  $\alpha$  is a local weighting factor that depends on the human visual characteristics of the coefficient value in the watermarking space. To get the local weighting factor, we use the Noise Visibility Function (NVF) based on the non-stationary Gaussian image model [6]. The NVF characterizes the local texture of the image ranging from 0 to 1, where it takes 1 for flat areas and 0 for highly texture regions. The NVF for the practically important case of the non-stationary Gaussian image model has the following form [6];

$$NVF(x, y) = \frac{1}{1 + \sigma_{c_o}^2(x, y)}, \quad (3)$$

where  $\sigma_{c_o}^2(x, y)$  denotes the local variance of the original image  $c_o$  in a window centered on the pixel with

coordinates  $(x, y)$ . As one can see, the NVF is inversely proportional to the local image energy defined by the local variance. In this paper, the local weighting factor  $\alpha$  is  $\log_{10}(1/NVF(x, y))$  or  $\sqrt{1/NVF(x, y)}$  depending on the applications at hand. In general, the added watermark  $w_a (= \alpha w_m)$  has zero-mean Gaussian distribution. Finally, the watermarked image is obtained by adding the added watermark pattern  $w_a$  to original image  $c_o$ .

Although the image is modified without the perceptual degradation of quality, the value  $\alpha$  is preserved because the local characteristics of the image do not change very much. Using this invariance of local weighting factor, in the detection process, the exact power of the embedded watermark can be calculated without original image. Then, the exact channel noise power, which improves the watermark detection performance in LDPC codes, can be calculated from the power of the embedded watermark and the estimated watermark.

### 3. Watermark Estimation Process

During transmission, the watermarked image  $c_w$  is corrupted by i.i.d zero-mean white Gaussian noise  $n$  with unknown standard deviation  $\sigma_n$ . Thus, at the receiver end, the noisy watermarked image can be expressed as follows;

$$c_{wn} = c_o + w_a + n = c_o + w_{an}. \quad (4)$$

We regard the sum of added watermark  $w_a$  and channel noise  $n$  as the corrupted watermark  $w_{an}$ . We can assume that the random variable of  $w_{an}$  has zero-mean white Gaussian distribution because the added watermark and the noise signal have the identical zero-mean white Gaussian distribution. Then, our goal is to estimate the added watermark with noise  $\hat{w}_{an}$  from the noisy watermarked image  $c_{wn}$  such that mean square error is to be minimum.

Let  $C_{wn}$  be the two-dimensional orthogonal discrete wavelet transform of  $c_{wn}$ . Then, the wavelet coefficients of  $C_{wn}$  have four sub-bands such as  $LL$ ,  $LH$ ,  $HL$  and  $HH$ . The sub-bands  $HH_k$ ,  $HL_k$  and  $LH_k$  contain the detailed components, where  $k \in \{1, 2, 3, \dots, K\}$  and  $K$  is the total number of decompositions. The sub-band  $LL_K$  is the low-resolution residue. The wavelet de-signaling method is executed to the coefficients of  $C_{wn}$  in the detail sub-bands with an adaptive thresholding values to obtain the estimated noisy watermark frequency  $\hat{W}_{an}$ . Then, the de-signaled estimate is inverse-transformed to obtain the estimated noisy watermark  $\hat{w}_{an}$ . This algorithm, which is very simple to implement and computationally efficient [2], can be summarized with the following steps.

- A. Perform multi-scale decomposition of the noisy watermarked image  $c_{wn}$  using the wavelet transform.
- B. Estimate the channel noise variance  $\sigma_n^2$  as follows [2].

$$\sigma_n^2 = \left[ \frac{\text{median}\{C_{wn}^{HH_1}(u, v)\}}{0.6745} \right]^2, \quad (5)$$

where  $C_{wn}^{HH_1}(u, v)$  is a wavelet coefficients with coordinate  $(u, v)$  in  $HH_1$  band.

- C. For each resolution scale, compute the scale parameter  $\beta_k$  [2].

$$\beta_k = \sqrt{\log\left(\frac{Q_k}{K}\right)}, \quad (6)$$

where  $Q_k$  is the total number of coefficients in each sub-band at  $k^{th}$  scale.

- D. For the coefficients in the  $k^{th}$  sub-band  $C_{wn}^{Sb_k}$ ,  $Sb_k \in \{HH_k, HL_k, LH_k\}$  (except the low-pass residue), compute the standard deviation of the sub-band under consideration  $\sigma_{C_{wn}^{Sb_k}}$ . Then, compute threshold  $T_{C_{wn}^{Sb_k}}$  as follows [2].

$$T_{C_{wn}^{Sb_k}} = \frac{\beta_k \sigma_n^2}{\sigma_{C_{wn}^{Sb_k}}} \quad (7)$$

For the  $k^{th}$  sub-bands, the threshold value is inversely proportional to the standard deviation of the sub-band under consideration  $\sigma_{C_{wn}^{Sb_k}}$ , because the scale parameter  $\beta_k$  depends upon the sub-band size and the parameter  $\sigma_n^2$  is the fixed value. That means, the more noise, the higher threshold value. This relationship results in separating noise signal from the noisy image adaptively.

- E. Apply thresholding value in (7) to each coefficient except the low-resolution residue as follows;

$$MC_{wn}^{Sb_k}(u, v) = \begin{cases} 0 & \text{if } |C_{wn}^{Sb_k}(u, v)| > T_{C_{wn}^{Sb_k}} \\ C_{wn}^{Sb_k}(u, v) & \text{otherwise} \end{cases} \quad (8)$$

- F. Invert the multi-scale decomposition to re-construct the de-signaled estimate  $\hat{w}_{an}$ .

Considering the watermark as noise in the watermarked image, the attacker may easily estimate the original watermark. This is often called estimation-based attacks such as watermark removal, remodulation and copy attack [6]. To be robust against the estimation-based attacks, we need to embed the watermark such that it is difficult to predict the watermark. This issue has been solved in [7] as Power-Spectrum Condition (PSC).

#### 4. LDPC Encoding and Decoding

In communication system, ECC definitely helps to achieve more reliable transmission. Since watermarking is also viewed as a communication system, we adopt LDPC codes as the ECC. That is, the encoder takes the message sequence ( $s_m$  in this paper) as input, adds controlled redundancy to it, and sends out a longer coded sequence ( $w_m$  in this paper). The decoder uses the redundancy introduced by the encoder to detect and correct errors that occurred during transmission.

LDPC coding is a special class of linear parity check coding. It needs a generator matrix  $G$  and a parity check matrix  $H$ , which satisfies  $Hw_m^T \bmod 2 = 0$  for all codewords  $w_m$ . The matrix  $H$  is defined as follows. Each column and row of  $H$  consists of small and identical number of 1. And the inner product between any two columns is not greater than 1. The matrix  $H$  is constructed at random subject to these constraints. The matrix  $G$  is generated from the matrix  $H$  by the relationship  $GH^T \bmod 2 = 0$ . Then, the message sequence vectors  $s_m$  is encoded by matrix  $G$  such that  $w_m = s_m G \bmod 2$  and is BPSK (Binary Phase Shift Keying) modulated such that  $\{0,1\} \rightarrow \{-1,1\}$ .

The decoding process is to find the most probable vector  $\tilde{w}_{an}$  such that  $H\tilde{w}_{an}^T \bmod 2 = 0$ . This starts by calculating the probability that  $r^{th}$  entry of the estimated noisy watermark  $\hat{w}_{an}(r)$  has value 0 or 1 such that [4]

$$f^{0/1}(r) = \frac{1}{1 + \exp(-2\hat{w}_{an}(r)/\sigma_n^2)} \quad (9)$$

To calculate the likelihood of  $\hat{w}_{an}$ , we need the actual noise power of channel  $\sigma_n^2$ . However, in real situation, it is rarely available. So, the best performance of LDPC codes may not be achieved. In this paper, we estimate the channel noise power  $\sigma_n^2$  from the estimated watermark  $\hat{w}_{an}$  with the known watermark power  $\sigma_{w_a}^2$ . Since the embedded watermark  $w_a$  has zero-mean distribution and the invariance of the local weighting factor between original image and noisy watermarked image is preserved, the power of the embedded watermark is given by

$$\begin{aligned} \sigma_{w_a}^2 &= E[w_a^2] = E[\alpha^2 w_m^2] \\ &\approx E[\hat{\alpha}^2 w_m^2] = E[\hat{w}_a^2] = \sigma_{\hat{w}_a}^2, \end{aligned} \quad (10)$$

where  $E[\cdot]$  is expectation operator and the value  $\hat{\alpha}$  is a local weighting factor from the noisy watermarked image  $c_{wn}$ . Although the watermarked image is corrupted, the value  $\hat{\alpha}$  is very similar to  $\alpha$  because the local characteristics of the image do not change very much.

The estimated noisy watermark  $\hat{w}_{an}$  with zero-mean distribution can be expressed by the channel noise estimate  $\hat{n}$  with zero-mean distribution and the embedded watermark  $\hat{w}_a$ . Then, the power of the estimated noisy watermark  $\sigma_{\hat{w}_{an}}^2$  can be expressed as follows;

$$\begin{aligned} \sigma_{\hat{w}_{an}}^2 &= E[\hat{w}_{an}^2] = E[(\hat{n} + w_a)^2] = E[\hat{n}^2 + 2\hat{n}w_a + w_a^2] \\ &= E[\hat{n}^2 + w_a^2] = \sigma_{\hat{n}}^2 + \sigma_{w_a}^2 \approx \sigma_{\hat{n}}^2 + \sigma_{\hat{w}_a}^2. \end{aligned} \quad (11)$$

Finally, using (11), we can calculate the power of the estimated channel noise as follows;

$$\sigma_{\hat{n}}^2 = \sigma_{\hat{w}_{an}}^2 - \sigma_{\hat{w}_a}^2. \quad (12)$$

To decode LDPC codes with the estimated noise power, we use the sum-product algorithm [8][9] that is sometimes referred to as "Belief Propagation (BP) algorithm" [10]. The sum-product algorithm passes message along the edges of a bipartite factor graph [11] that describes the conditional joint probability mass function of the codeword symbols given the received channel output.

To extract the message after LDPC decoding, we apply an M-ary demodulator to LDPC decoded sequence  $\tilde{s}(l)$  by calculating the normalized correlation as follows;

$$m^* = \arg \max_{m \in \{1, 2, \dots, 2^{A/B}\}} \frac{\sum_{l=1}^L \tilde{s}(l) s_m(l)}{\sqrt{\sum_{l=1}^L \tilde{s}^2(l)} \sqrt{\sum_{l=1}^L s_m^2(l)}}, \quad (13)$$

where  $s_m(l)$  is a sequence with index  $m$ . We choose the optimal index  $m^*$  of the maximum correlated sequence as the embedded sequence. This index  $m^*$  represents the embedded character.

#### 5. Experimental Results

We first partition the given image into  $B$  blocks of size 2048 (i.e.  $R = 2048$ ). The sequence used in M-ary modulation is the m-sequence of length 1024 (i.e.  $L = 1024$ ) and 8 bits are embedded into each block. The window size in NVF is  $5 \times 5$ . The wavelet transform used in this experiments is a three level decomposition with Daubechies-8 filters. The dimensions of the matrices used in LDPC codes are as follows.

An  $(2048, 3, 6)$  LDPC code with rate  $CR = 1/2$  is used in our experiments.

Message sequence :  $m_s = 1 \times L = 1 \times 1024$

Codeword :  $w_m = 1 \times R = 1 \times 2048$

Generator matrix :  $G = L \times R = 1024 \times 2048$

Parity check matrix :  $H = (R - L) \times R = 1024 \times 2048$

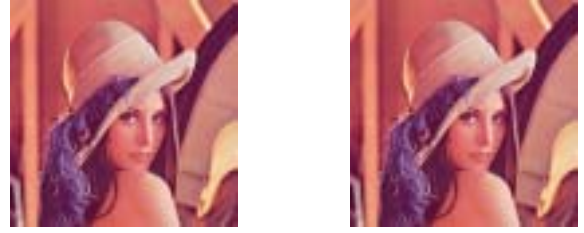
(14)

The test images are Lena, Baboon and Fishing Boat etc. The original and watermarked images are shown in Fig. 2. All the watermarked images have the PSNR about 40dB compared to the original ones and the watermarks are invisible. Figure 3 shows that there is little difference between the original watermark ( $w_a$ ) and the one ( $w_{an}$ ) after JPEG compression (compression factor 50 in Jasc Paint Shop Pro™ ver. 7.0). This invariance of the added watermarks enables us to calculate the exact power of the embedded watermark in detection process. That is, we can calculate the exact power of the channel noise from the estimated watermark to improve the watermark decoding performance in LDPC codes. Figure 4 shows the difference between the original watermark and the estimated watermark by adaptive thresholding estimation method in case of no corruption in the image. From this experimental result, we can confirm that the estimated watermark is very similar to the original one. Figure 5 depicts the difference between the original noise and the estimated one in case where the watermarked image is compressed by compression factor 50 showing no significant difference between them. Finally, figure 6 shows the watermark decoding performance in LDPC codes between original, estimated, and assumed channel noise power ( $\pm 0.5\text{dB}$ ). As one can see in graph, the estimated noise power proposed in this paper yields almost similar BER (Bit Error Rate) with those of known noise power.

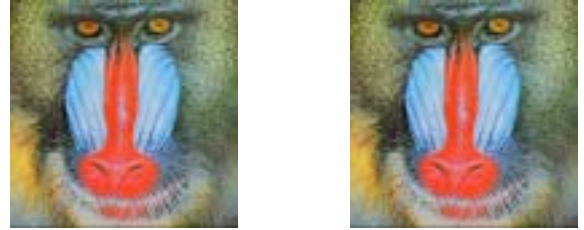
## 6. Conclusions

In this paper, we improved the watermarking performance by estimating the watermark from the attacked image and applying LDPC codes to the estimated watermark. The watermark estimation can be done by an adaptive thresholding estimation method for image denoising in the wavelet domain based on the generalized Gaussian distribution modeling of sub-band coefficients. When LDPC codes are applied to the estimated watermark, we can solve the mismatch problem between the assumed noise power and the true one by calculating the noise power from the estimated watermark. Simulation results demonstrate the superior performance of the proposed algorithm.

The proposed algorithm can be extended to be robust against the geometrical attacks such as rotation and scaling etc. To this end, a "self-reference" scheme can be employed by exploiting the periodicity of the repeating block-wise watermarks [12][13].



(a) Original and watermarked Lena image  
(PSNR : 39.2859)



(b) Original and watermarked Baboon image  
(PSNR : 40.1552)



(c) Original and watermarked Fishing Boat image  
(PSNR : 39.5781)

Fig. 2 Original image and watermarked image by proposed algorithm

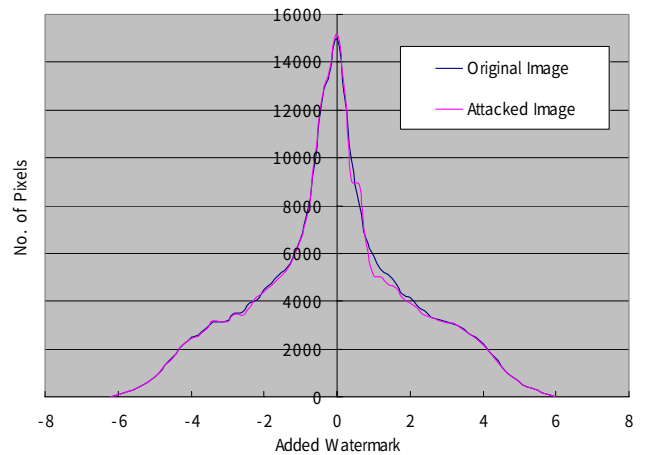


Fig. 3 Distribution of added watermark between original and attacked image.

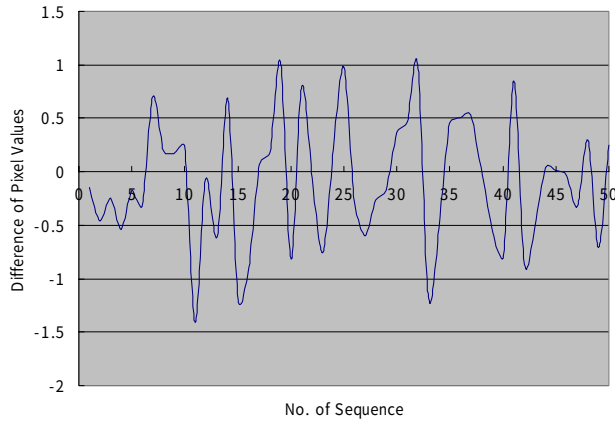


Fig. 4 Difference of the true added watermark and the estimated watermark

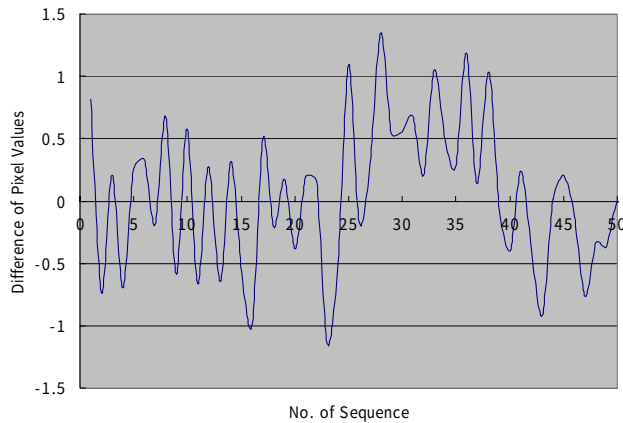


Fig. 5 Difference of the true channel noise and the estimated noise

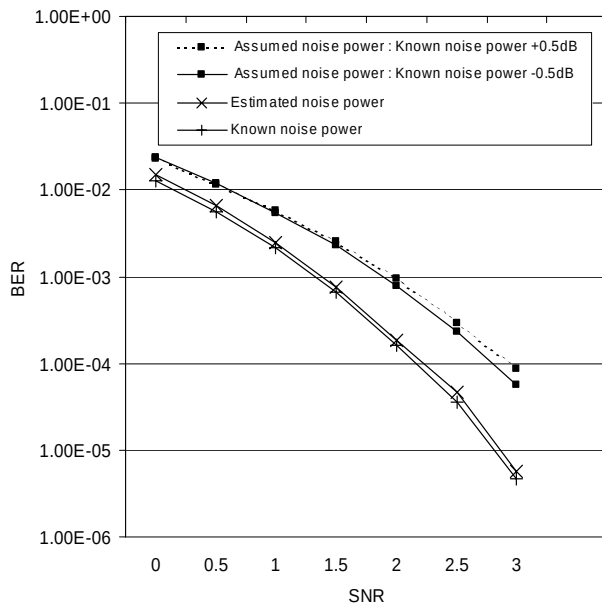


Fig. 6 BER curve between original noise, estimated and assume noise ( $\pm 0.5\text{dB}$ ).

## 7. References

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